

and a simple prediction algorithm head-to-head in a horse race, and determine which can more accurately classify the extent of a patient's brain impairment. As before, the model utilizes a systematic approach based on a statistical model of prior data; meanwhile, the humans can utilize their vast expertise and intuition based on years of experience.

The results from the study should be, by now, somewhat expected, yet they are still striking. The simple quantitative model has a classification accuracy ratio of 83.3 percent, which was much higher than for the experienced clinicians, who had a success rate of only 58.3 percent. Interestingly, the inexperienced clinicians were slightly better at 62.5 percent. As we saw previously, the model (and the novices) clearly beat the experts.

But this time the researchers took their analysis one step further. They wanted to explore what would happen when the experts were armed with the powerful prediction model. A natural hypothesis is that experts, equipped with the model, would outperform the stand-alone model. In other words, models might represent a floor on performance, to which the experts could add performance.

In follow-on tests, the researchers gave the experts the output of the model and disclosed that the model had “previously demonstrated high predictive validity in identifying the presence or absence of intellectual deterioration associated with brain damage.” Using the model, experienced clinicians significantly improved their accuracy ratio from 58.3 percent to 75 percent and the inexperienced clinicians moved from 62.5 percent to 66.5 percent. Nonetheless, the experts were still unable to outperform the stand-alone model, which had already established the gold standard 83.3 percent success rate. This study suggests that models don't represent a floor on performance; rather, models reflect a ceiling on performance, from which the experts detract performance. The “secret sauce” of human judgment ruins the beautiful simplicity and effectiveness of a mechanistic calculation (see [Figure 2.3](#)).